

Official Engineering Backlog

Project NOVA Master Planning & Sprint Backlog

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Executive Summary

This document is the single source of truth for all future engineering implementation. It maps the Project NOVA architectural blueprints into a structured, executable 4-Phase agile backlog. Every task defines rigid dependencies and acceptance criteria to ensure the orchestration Kernel, security boundaries, and cognitive engines are constructed deterministically.

Phase 1: Platform Foundation

Focus: Establishing the orchestration kernel, core abstractions, and strict security perimeters before any AI models or external OS integrations are permitted.

Epic 1: Engineering Governance

Establishing the rules of engagement for the engineering team.

Task ID: `TSK-101`

- **Title:** Establish Official Repository Specification
- **Description:** Draft and approve NOVA-REP-002 defining the exact Python module boundaries, ``01_Source/`` directory policing, and plugin storage locations.
- **Dependencies:** None
- **Priority:** CRITICAL
- **Estimated Complexity:** Low (Documentation)
- **Acceptance Criteria:** ``NOVA-REP-002`` compiled to PDF and merged into ``ARCHITECTURE_STATUS.md``.
- **Status:** ``TODO``
- **Suggested Sprint:** Sprint 1

Task ID: ``TSK-102``

- **Title:** Establish Technology Stack Specification
- **Description:** Draft and approve NOVA-TECH-001 locking in Python 3.12, Pytest, Pydantic, and all Provider choices (Vector DBs, Whisper, Tesseract) with justifications.
- **Dependencies:** ``TSK-101``
- **Priority:** CRITICAL
- **Estimated Complexity:** Medium (Research & Documentation)
- **Acceptance Criteria:** ``NOVA-TECH-001`` compiled and merged.
- **Status:** ``TODO``
- **Suggested Sprint:** Sprint 1

Epic 2: Core Frameworks Implementation

Building the abstract interfaces that govern system execution.

Task ID: ``TSK-103``

- **Title:** Implement Capability Framework Base Models
- **Description:** Code the ``ICapability`` abstract base class and ``CapabilityMetadata`` pydantic schemas. Define the standard ``start()``, ``execute()``, ``stop()`` lifecycle.
- **Dependencies:** Kernel Bootstrap (Completed)
- **Priority:** HIGH
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** Base classes exist in ``nova.core.capabilities``. Pytest passes for schema validation.
- **Status:** ``TODO``
- **Suggested Sprint:** Sprint 2

Task ID: ``TSK-104``

- **Title:** Implement Provider Abstraction Layer (PAL)
- **Description:** Code the generic interfaces for ``ILLMProvider``, ``IVisionProvider``, and ``IVoiceProvider`` to ensure hot-swappable AI backends.
- **Dependencies:** ``TSK-103``

- **Priority:** HIGH
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** Abstract base classes defined in `nova.core.providers` with zero concrete dependencies (no OpenAI/Anthropic SDKs imported).
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 2

Task ID: `TSK-105`

- **Title:** Implement Plugin Discovery & Manifest System
- **Description:** Build the `PluginRegistry` to safely scan `.json` manifests in the `plugins/` directory, verifying cryptosignatures before loading Python modules into memory.
- **Dependencies:** `TSK-103`
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** System successfully discovers and instantiates a mock "Hello World" plugin dynamically.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 2

Epic 3: System Infrastructure & Security

Fortifying the execution pipeline.

Task ID: `TSK-106`

- **Title:** Implement Configuration & Secrets Manager
- **Description:** Extend the configuration loader to securely read API keys (OpenAI, etc.) from isolated `.env` vaults and inject them only into authorized Providers.
- **Dependencies:** `TSK-102`
- **Priority:** HIGH
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** Keys are injected securely without ever being written to standard output logs.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 3

Task ID: `TSK-107`

- **Title:** Implement Permission & Sandbox Framework
- **Description:** Build the `PermissionManager`. Intercept all requests from Capabilities to the OS, ensuring the `ExecutionToken` has the required scopes (e.g., `fs.read`, `process.spawn`).
- **Dependencies:** `TSK-105`
- **Priority:** CRITICAL
- **Estimated Complexity:** High
- **Acceptance Criteria:** A mock malicious plugin attempting to access `C:\Windows` throws a `PermissionDeniedError` and is forcibly unloaded.

- **Status:** `TODO`
- **Suggested Sprint:** Sprint 3

Task ID: `TSK-108`

- **Title:** Implement Global Event Framework
- **Description:** Expand the Kernel's `AsyncioEventBus` to handle multi-threaded asynchronous publishing, dead-letter queuing, and strict schema validation for cross-capability communication.
- **Dependencies:** Kernel Bootstrap (Completed)
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** Event bus routes 10,000 simulated events per second without memory leaks.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 3

Task ID: `TSK-109`

- **Title:** CI/CD & Platform Review Integration
- **Description:** Establish GitHub Actions / local CI pipelines enforcing `flake8`, `black`, `mypy`, and 90% test coverage before any merge to main. Write developer onboarding docs.
- **Dependencies:** All Sprint 3 tasks.
- **Priority:** MEDIUM
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** CI pipeline triggers successfully on mock PR.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 4

Phase 2: Platform Capabilities

Focus: Bridging NOVA into the physical OS environment through strictly sandboxed, single-responsibility capabilities.

Epic 4: Native Capabilities

Task ID: `TSK-201`

- **Title:** Implement Desktop Automation Capability
- **Description:** Code the concrete implementation of `ICapability` wrapping PyAutoGUI and OS-level window management APIs (Windows User32).
- **Dependencies:** `TSK-103`, `TSK-107`
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** NOVA can programmatically locate and focus a specific window, and dispatch a mock mouse click.

- **Status:** `TODO`
- **Suggested Sprint:** Sprint 5

Task ID: `TSK-202`

- **Title:** Implement Vision & Screen Understanding Capability
- **Description:** Code the screenshot buffering mechanism and OCR/Bounding Box extraction.
- **Dependencies:** `TSK-201`
- **Priority:** HIGH
- **Estimated Complexity:** Very High
- **Acceptance Criteria:** Takes a screenshot, extracts UI elements to a structured JSON tree without invoking LLM logic.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 5

Task ID: `TSK-203`

- **Title:** Implement Browser Automation Capability
- **Description:** Code the concrete wrapper around Playwright for DOM parsing, DOM manipulation, and JavaScript injection.
- **Dependencies:** `TSK-103`
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** Boot a headless Chromium instance, navigate to a URL, and extract the accessibility tree.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 6

Task ID: `TSK-204`

- **Title:** Implement Voice Processing Capability
- **Description:** Code the STT (Speech-to-Text) microphone listener buffer, VAD (Voice Activity Detection), and TTS playback queue.
- **Dependencies:** `TSK-104`
- **Priority:** MEDIUM
- **Estimated Complexity:** High
- **Acceptance Criteria:** Successfully records 5 seconds of local audio, detects silence, and converts text to synthesized speech output.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 6

Task ID: `TSK-205`

- **Title:** Implement Persistent Memory Capability
- **Description:** Initialize SQLite (Semantic metadata) and local VectorDB (embeddings). Implement chunking and retrieval interfaces.
- **Dependencies:** `TSK-104`

- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** Successfully stores a text string as an embedding and retrieves it via cosine similarity search.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 7

Task ID: `TSK-206`

- **Title:** Implement Workflow Capability
- **Description:** Implement multi-step Macro recording and playback mechanism (triggering sequences of Desktop/Browser events).
- **Dependencies:** `TSK-201`, `TSK-203`
- **Priority:** MEDIUM
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** Successfully records a 3-step action list and replays it blindly.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 7

Phase 3: Intelligence

Focus: Connecting the cognitive LLM engines to the execution Capabilities via the Event Bus.

Epic 5: Cognitive Engines

Task ID: `TSK-301`

- **Title:** Implement Context & Observation Engine
- **Description:** A service that subscribes to Vision/Browser events, compresses the state (DOM/Screen), and formats it into an LLM-digestible prompt buffer.
- **Dependencies:** `TSK-202`, `TSK-203`
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** Generates a unified JSON representation of the current OS state.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 8

Task ID: `TSK-302`

- **Title:** Implement Primary Reasoning Planner
- **Description:** The core AI service that takes User Intents (from Voice/Text) + Context Buffer, and outputs JSON tool-calls mapped to the active Capability Registry.
- **Dependencies:** `TSK-301`, `TSK-105`
- **Priority:** CRITICAL

- **Estimated Complexity:** Very High
- **Acceptance Criteria:** Planner accurately outputs a valid `DesktopCapability.Click` payload when asked to "Click the start button".
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 8

Task ID: `TSK-303`

- **Title:** Implement Reflection & Learning Loop
- **Description:** An asynchronous evaluator that analyzes whether the Planner's previous action successfully changed the OS state as intended. Writes successful paths to the Memory Capability.
- **Dependencies:** `TSK-302`, `TSK-205`
- **Priority:** HIGH
- **Estimated Complexity:** High
- **Acceptance Criteria:** If an action fails, Planner automatically adjusts trajectory on the next tick without user intervention.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 9

Phase 4: Applications

Focus: The presentation layer directly facing the end-user.

Epic 6: Presentation Layer

Task ID: `TSK-401`

- **Title:** Develop NOVA CLI Interface
- **Description:** Create a rich terminal UI (using Textual or Rich) for developers to interact with the system without a heavy GUI.
- **Dependencies:** `TSK-108`
- **Priority:** HIGH
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** User can type commands into terminal and view streamed Event Bus logs natively.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 10

Task ID: `TSK-402`

- **Title:** Develop Desktop GUI (System Tray & Overlay)
- **Description:** A lightweight desktop application (PyQt or Web-UI) that runs in the background, providing a transparent overlay for bounding boxes and a microphone hotkey.
- **Dependencies:** `TSK-204`, `TSK-302`

- **Priority:** MEDIUM
- **Estimated Complexity:** High
- **Acceptance Criteria:** User presses `Ctrl+Space` to wake NOVA, overlay displays visual feedback of AI perception.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 11

Task ID: `TSK-403`

- **Title:** REST/WebSocket API & External Deployments
- **Description:** Expose the Kernel Event Bus over authenticated WebSockets to allow future Mobile or Web apps to communicate with the local instance.
- **Dependencies:** `TSK-107`
- **Priority:** LOW
- **Estimated Complexity:** Medium
- **Acceptance Criteria:** External client successfully authenticates and triggers a macro remotely.
- **Status:** `TODO`
- **Suggested Sprint:** Sprint 12